UNTRM AI WORKSHOP · SESSION 3

Coding with a Copilot: Adapting Pre-Trained Models for Your Research

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This session's plan

PART 1 · 40 MIN

Reusing models when data is scarce

Transfer learning, foundation models, VLMs, augmentation

PART 2 · 20 MIN

AI code assistants

Getting started with GitHub Copilot

PART 3 · 60 MIN

Use cases by prompting

The experiment ladder on bean leaves

Part 1 gives the ideas; Parts 2 and 3 turn them into code with a copilot.

PART 1

Reusing models when data is scarce

Transfer learning, foundation models, vision-language models and data augmentation

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THE PROBLEM

What if you only had 25 leaves per class?

1,034

training photos in the beans dataset we used yesterday

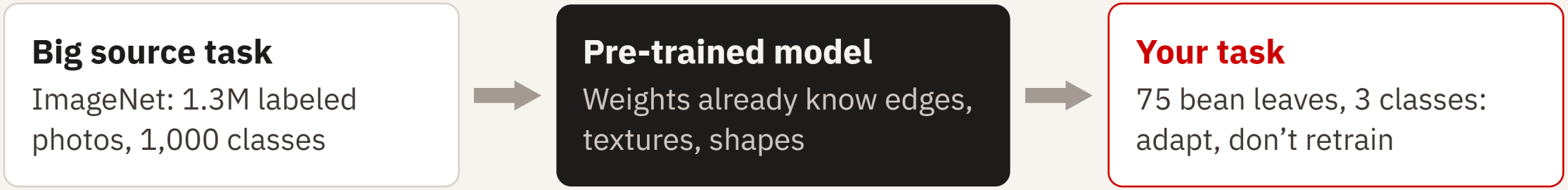
75

what a new crop, pest or microscope often starts with: 25 per class

Expert labels are slow and expensive. Training a deep network from scratch on 75 images will overfit.

TRANSFER LEARNING

Reuse, don’t restart



From scratch: random weights, lots of data, hours to days of GPU time.

Reuse: start from learned weights, adapt in minutes on a free GPU.

WHY IT WORKS

Early layers are general, late layers are specific

Layer 1	Layers 2–3	Middle	Last layer
Edges	**Textures**	**Parts**	**Objects**
Lines, color blobs	Spots, veins, grain	Leaf margins, lesions	“Dog”, “car” …

Transfer: useful for almost any image | **Replace**

Keep the general layers; swap the last one for your own classes.

TWO STRATEGIES

Feature extraction vs. fine-tuning

Feature extraction



Train only the new head. Fast, safe with very little data.

Fine-tuning



Also update the top layers, slowly. Better fit, needs more data.

Start with feature extraction; fine-tune only if validation says it helps.

IN PRACTICE

Five habits for fine-tuning

1. **Match the input:** same image size and normalization the model was trained with
2. **Start frozen:** train the new head first
3. **Go slow:** a smaller learning rate for the pre-trained layers
4. **Unfreeze gradually:** top blocks first, deeper only if it helps
5. **Watch validation:** stop early when it stops improving

CHOOSING A STRATEGY

How much data, and how similar?

	Similar to the source	Different from the source
Small data	**Feature extraction** Bean leaf photos	**Earlier layers or a domain model** Spore micrographs
Large data	**Fine-tune top layers** Drone images of crops	**Fine-tune deeply** Proteins: start from a protein model

OTHER WAYS TO TRANSFER

Borrow from other data and other tasks

Across datasets

Train on a large public set, then adapt to your farm or your microscope.

From unlabeled data

Pre-train an autoencoder on all your images, then train a classifier on the few labeled ones.

Multi-task

One backbone, two heads: disease type and severity learn from each other.

The unlabeled-data idea, at a huge scale, is how foundation models are built.

WATCH OUT

Four ways transfer learning goes wrong

⚠ **Domain mismatch**
Lab photos ≠ field photos

⚠ **Catastrophic forgetting**
Too high a learning rate erases what it knew

⚠ **Licenses**
Check weights and data terms before you publish

⚠ **Test-set leakage**
The model may have seen your “new” images

Pre-trained model vs. foundation model

Aspect	Classic pre-trained (e.g., ResNet)	Foundation (e.g., CLIP, DINOv2)
Training data	~1.3M labeled photos, 1,000 classes	Hundreds of millions of images, image–text pairs or sequences
Training signal	Supervised: human labels	Self-supervised: the data is its own label
Out of the box	Needs a new head and training	Zero-shot by text, general embeddings, promptable

Broader, bigger, and less transparent

Aspect	Classic pre-trained	Foundation
Scope	One data type, one task family	Many tasks; some mix image and text
Adapting it	Fine-tune the network	Prompt, embeddings, LoRA, or full fine-tune
Size and cost	Millions of parameters; runs on a laptop	Up to billions; GPU or paid API
Risks	Known, documented data	Web bias, unclear data sources, licenses, confident errors

SELF-SUPERVISED LEARNING

How they learn without labels

Fill in the blank

Hide patches of an image, or words, or amino acids, and predict them.

LLMs, ESM-2, MAE

Match image and caption

Pull each photo close to its own caption, away from the others.

CLIP, BioCLIP

Compare two views

Two crops of one image should give the same features.

DINOv2

USING A FOUNDATION MODEL

Climb only as high as you need

1

Prompt / zero-shot

No training. Describe the classes in words.

2

Embeddings + linear

Frozen model; train a tiny classifier.

3

LoRA, adapters

Train small add-on weights.

4

Full fine-tune

Update every weight.

Less data and compute → More data, GPU, risk

FOR YOUR FIELDS

Foundation models you can download today

CLIP · DINOv2
General images: zero-shot and embeddings

BioCLIP
Organisms: trained on TreeOfLife-10M

ESM-2
Proteins: embeddings, structure (ESMFold)

SAM
Segment anything: Session 4

Most are free on Hugging Face; check each license.

THE TAKEAWAY

With tens of images per class, start with a foundation model's embeddings and a simple classifier.

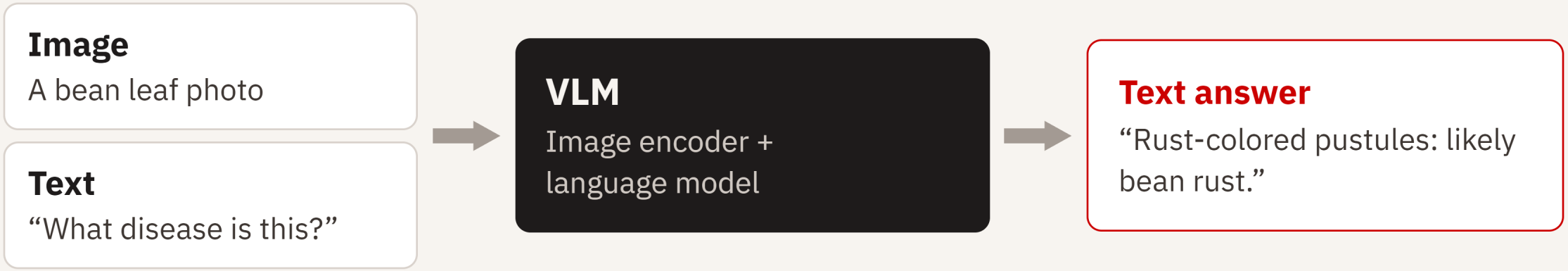
Fine-tune only when that baseline is not enough.

But a classic pre-trained CNN (ResNet, MobileNet) can be the better choice when:

- It must run offline, on a phone or edge device
- You need fast, cheap inference at scale
- You have hundreds of labels to fine-tune
- Licenses or a known training set matter

VISION-LANGUAGE MODELS (VLMS)

Ask a question about an image, in words



Examples: Gemini, GPT-4o, Qwen-VL (in an app or API); PaliGemma, Florence-2 (open, run in Colab). The “image agent” from Session 1 was a VLM.

LLM VS. VLM

A VLM is an LLM that learned to read images



Aspect	LLM (e.g., ChatGPT text)	VLM (e.g., Gemini with a photo)
Input	Text only	Text plus images (some: video)
Training	Next-word prediction on text	Plus image–caption pairs
Good at	Writing, summarizing, code	Questions about images
Images?	Via tools: OCR, a trained visual model	Directly, inside the model

THE PROMPT IS PART OF THE MODEL

Same photo, three prompts, three tasks

Focus	“Ignore the background. Describe only the lesions on the central leaf.”
Classify	“Label it healthy, angular leaf spot or bean rust. Answer in JSON with a confidence. Say ‘unclear’ if you cannot tell.”
Measure	“Using the scale bar, estimate the diameter of the largest lesion.”

The prompt decides what the model looks at and what it returns.

VLMS IN RESEARCH

Where a VLM saves you time

Zero-shot labeling
A first answer with no training data

Pre-annotation
The model drafts, the expert corrects

Image quality control
Flag blurry, empty or mislabeled images

Reading figures
Gels, plots, instrument screens, field notes

A vision tool inside an agent
Specify, co-design, evaluate: the same skills as Session 1

WATCH OUT

A fluent answer is not a correct answer

- **Hallucinated details** that are not in the image
- **Weak counting** and measurement
- **Small details lost** when the image is downsized
- **Different answers** from run to run
- **Your data leaves** when you use an API
- **Evaluate it** like any classifier, on a labeled test set

DATA AUGMENTATION

Make new examples that keep the label

Images

Flips, rotations, crops, brightness, blur

Time series

Small shifts, scaling, added sensor noise

DNA

Reverse complement, if the biology allows

The test: would an expert still give it the same label?

SHOW OF HANDS

Safe augmentation, or not?

A Rotate a micrograph of fungal spores

B Shift the hue of a bean leaf photo

C Flip a drone image of a field

D Reverse a DNA sequence without complementing it

Hands up for “safe”. Then: what could go wrong?

GENERATIVE AUGMENTATION

Synthetic images for rare classes, with care

Diffusion models learn to remove noise step by step; started from pure noise and guided by text or an image, they create new images.

1 · Generate
Text-to-image, image-to-image or inpainting

→

2 · Filter
An expert rejects wrong ones

→

3 · Mix
A limited share of the training set

→

4 · Evaluate
On real test images only

Alternative: simulation and digital twins, when the physics is known (e.g., crop growth, sensor models).

REGULARIZATION

Three more guards against overfitting

Weight decay

Penalize large weights, so the model stays simple

Dropout

Randomly switch off neurons while training

Early stopping

Stop when validation error starts rising

⚠ **Augment the training set only, never validation or test.**

PUTTING IT TOGETHER

A recipe for small datasets

1. **Zero-shot check:** how far do CLIP or a VLM get with no training?
2. **Baseline:** frozen embeddings + a linear classifier
3. **Fine-tune the top** with augmentation and early stopping
4. **Go deeper** (more layers, LoRA) only if the data allows

Always: hold out whole groups (field, plant, slide) and report the spread across runs.

PART 2

AI code assistants

Getting started with GitHub Copilot

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COMING UP IN PART 3

beans128.npz

The experiment ladder: 25 leaves per class

Start with no training, then climb. Same 128 test images. Which step gives the best accuracy for the effort?

a	Prompt a VLM with the three labels: no training, no code
b	CLIP zero-shot: no training images at all
c	Small CNN from scratch
d	Frozen pre-trained backbone + linear classifier
e	Fine-tune the top layers + augmentation
f	**Optional:** add regularization (dropout, weight decay) and tune a model

Data: Makerere AI Lab “beans”, 128×128 copy · beans128.npz + README.md